INTRODUCTION

- **Concrete the man made rock is the most widely used construction material.**
- **More than 80,000 crores cum of concrete are poured every year.**
- **India has an installed capacity of 200 million tonnes of cement.**
- **Some Structures are designed with life of 100 years.**

So it is necessary that the concrete used is durable.

Durability

Definition –

- A durable concrete is one that performs satisfactorily in the working environment during its anticipated exposure conditions during service (IS 456-2000)
- Durability of concrete is its ability to resist weathering action, chemical attack, abrasion or any other process of deterioration (American Concrete Institute).
- When exposed to environment durable concrete is likely to retain its original form, quality and serviceability during its lifetime.
- Durable Concrete envisage limits for maximum water cement ratio, minimum cement content, cover thickness, type of cement used and presence of amount of chloride and sulphates in concrete. (IS-SP-28)
- As Low permeability as possible under situation. (IS-SP-23)

Consequences of Improper Quality and Inadequate Durability

- Loss of strength of concrete
- Concrete liable to be easily affected by deterrents
- Corrosion of rebars
- Loss of serviceability
- Unpleasant appearance
- Danger to persons and property
- Expensive repair costs
- Poor perception of concrete as a material
- Poor perception of agencies involved
- Reduction of service life
- External agencies like weathering, attack by natural or Industrial liquids, Gases, bacterial Growth etc.
- Alkali- aggregate reaction.
- Ingression of moisture/air facilitating corrosion of steel and cracking concrete cover.

DURABILITY

- Defined as its resistance to deterioration processes that may occur as a result of interaction with its environment (external) or between the constituent materials or their reaction with contaminants present (internal).

- Ability to with stand the damaging effects of the environment over a long period of time.

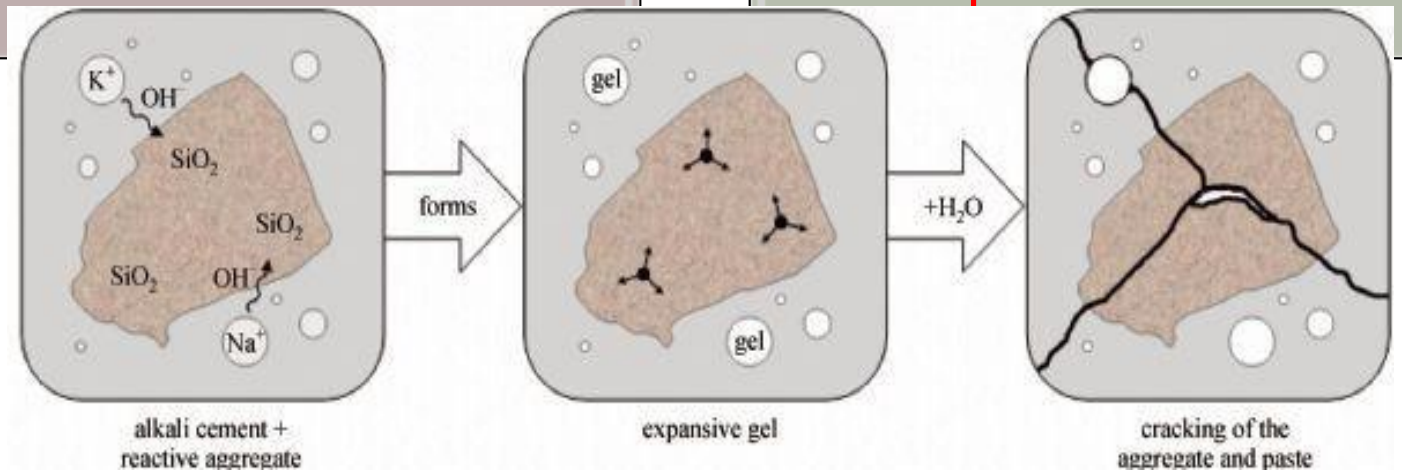
- The absence of durability maybe caused either by the environment to which the concrete is exposed i.e. external or internal causes.

INTERNAL CAUSES

- The alkali-aggregate reaction, volume changes due to the differences in thermal properties of aggregate and cement paste and the permeability of the concrete.

EXTERNAL CAUSES

- physical, chemical and mechanical
- weathering, occurrence of extreme temperature, abrasion, electrolytic action.
- The common forms of chemical attack : leaching out of cement and action of sulphates



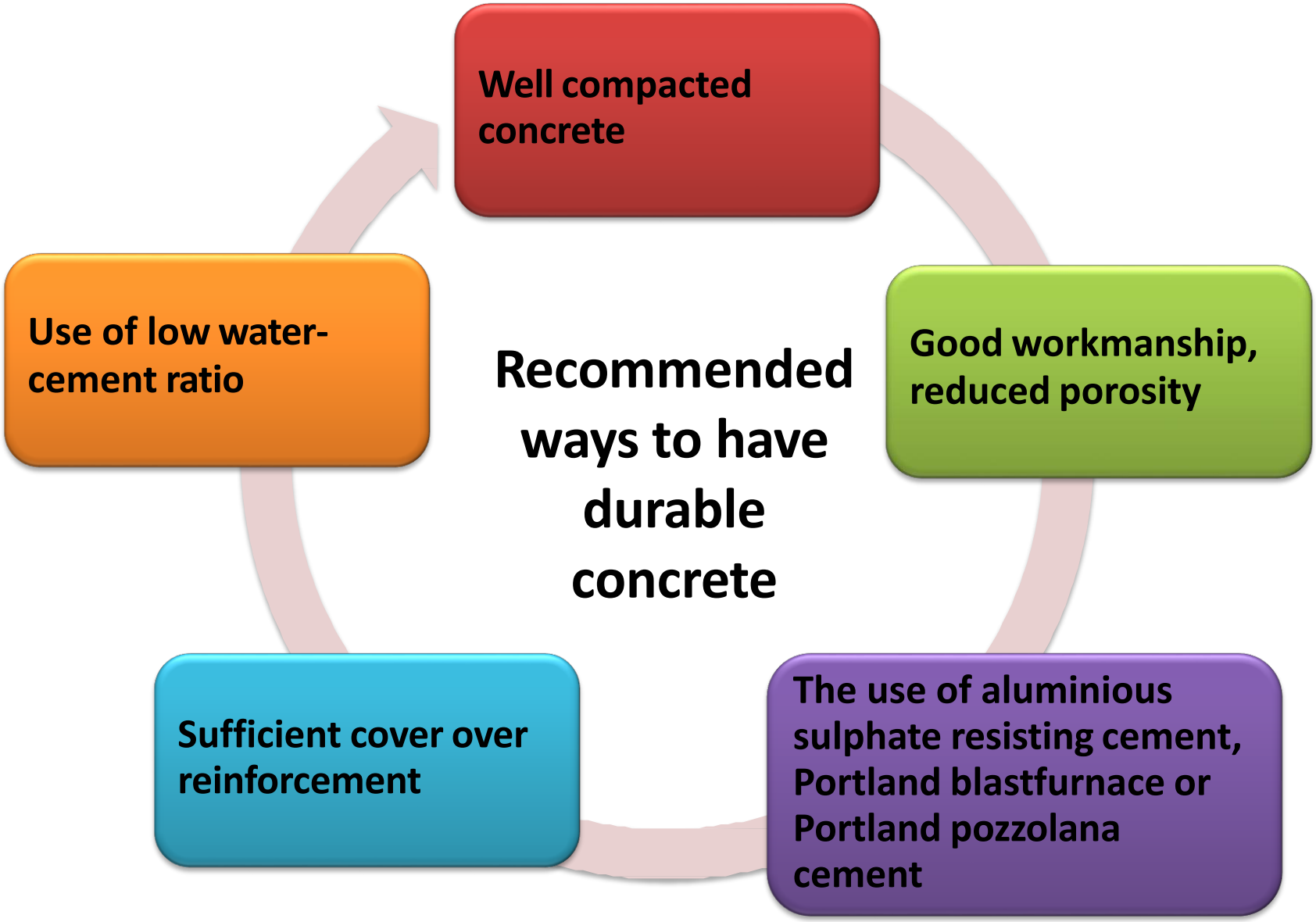
Leaching is the dissolution of some concrete components in a liquid. For example, lime is dissolved according to this reaction : Ca(OH)2 > Ca2+ + 2 OH- After leaching, concrete porosity is increased and its deterioration is enhanced

LEACHING IN CONCRETE








Recommended ways to have durable concrete
Well compacted concrete
Good workmanship, reduced porosity
The use of aluminious sulphate resisting cement, Portland blastfurnace or Portland pozzolana cement
Sufficient cover over reinforcement
Use of low water-cement ratio

Consequences of Inadequate Durability



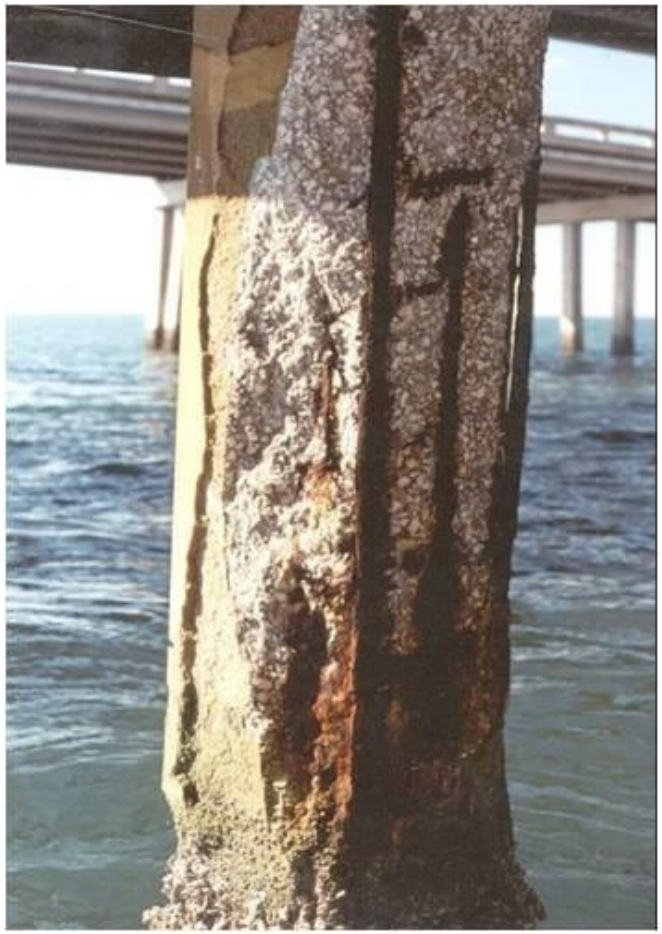


Consequences of Inadequate Durability





Consequences of Inadequate Durability





Factors Influencing Durability of Concrete (IS: 456-2000)

- The Environment
- Type and quality of constituent materials
- Cement content and W/C ratio of concrete
- Workmanship especially in compaction curing – it is very important
- Cover to embedded steel
- Shape and size of the member

Factors Affecting Durability

Durability depends on two main factors

a) The concrete system &

b) The service environment

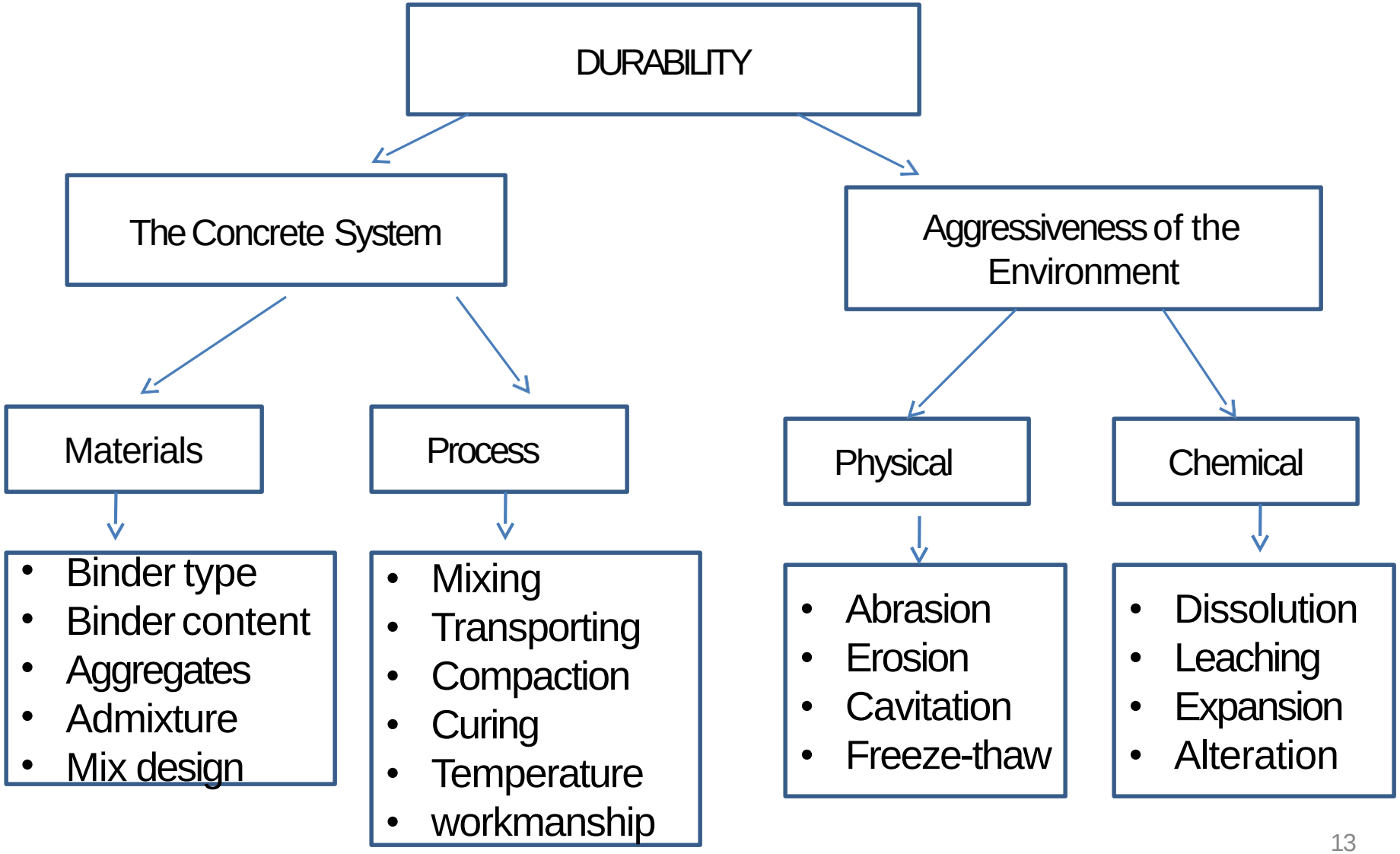
a) Concrete system is based on

- Quality and quantity of materials used and
- Processes involved in manufacture of concrete.

b) Service environment affects concrete by way of

- Physical actions and
- Chemical actions on concrete.

Factors Affecting Durability



Selection of Good Quality Materials
(Conforming to relevant IS codes)

- Cement (from Reputed Manufacturers)
- Sand (River / Crushed, Silt < 5%)
- Aggregates (Cubical in shape, Innocuous)
- Water (Tested) with PH value ranging 6 to 8
- Admixture (From Reputed Manufacturer)
- Compatibility of cement and plasticizer (PC based or Naphtha based) and 3rd generation superplasticiters of Polycarboxylates base, Polyacrylates based or Monovinyl alcohols based

UNSOUND MATERIALS

- Cement or aggregate is considered unsound when they cause unacceptable volume change, hardened concrete or mortar which causes cracks and affects durability.
- Aggregates containing certain materials such as shale, clay lumps, coal, iron pyrites etc show unsoundness later when concrete undergoes wetting and drying or freezing and thawing.
- More moisture absorption in aggregate (CA_1 or CA_{11}) is often used as a rough index for unsoundness
- Cement parameters that impact soundness are – free lime, MgO & excess gypsum

Water - Permissible Limits for Solids. IS 456:2000 Table-1

Sr. No.		Tested as per	Permissible Limit, Max
i)	Organic	IS 3025 (Part 18)	200 mg/l
ii)	Inorganic	IS 3025 (Part 18)	3000 mg/l
iii)	Sulphates (as SO_3)	IS 3025 (Part 24)	400 mg/l
iv)	Chlorides (as Cl)	IS 3025 (Part 32)	2000 mg/l for concrete not containing embedded steel and 500 mg/l for reinforced concrete work
v)	Suspended matter	IS 3025 (Part 17)	2000 mg/l

Impact of W/C Ratio On Durability

- Permeability is the contributory factor for volume change and higher W/C ratio is the fundamental cause of higher permeability.
- Use of higher W/C ratio – permeability – volume change – cracks – disintegration – failure of concrete is a cyclic process in concrete.
- For a durable concrete, use of lowest possible W/C ratio is the fundamental requirement to produce dense and impermeable concrete.
- Modern superplasticizers of Polymer base are so efficient that it is now possible to make flowing concrete with a W/C as low as 0.31 or even as low as 0.29 with increased slump more than 250mm.

Workmanship for Durable Concrete

- Batching
- Mixing
- Transportation
- Placing
- **Compaction**
- **Finishing**
- **Protection**
- **Curing**

Compaction of Concrete

- Ensuring suitable workability employing appropriate placing and compaction equipment
- Adequate compaction without segregation
- 1% voids reduces strength by 5%

Criticality of Curing

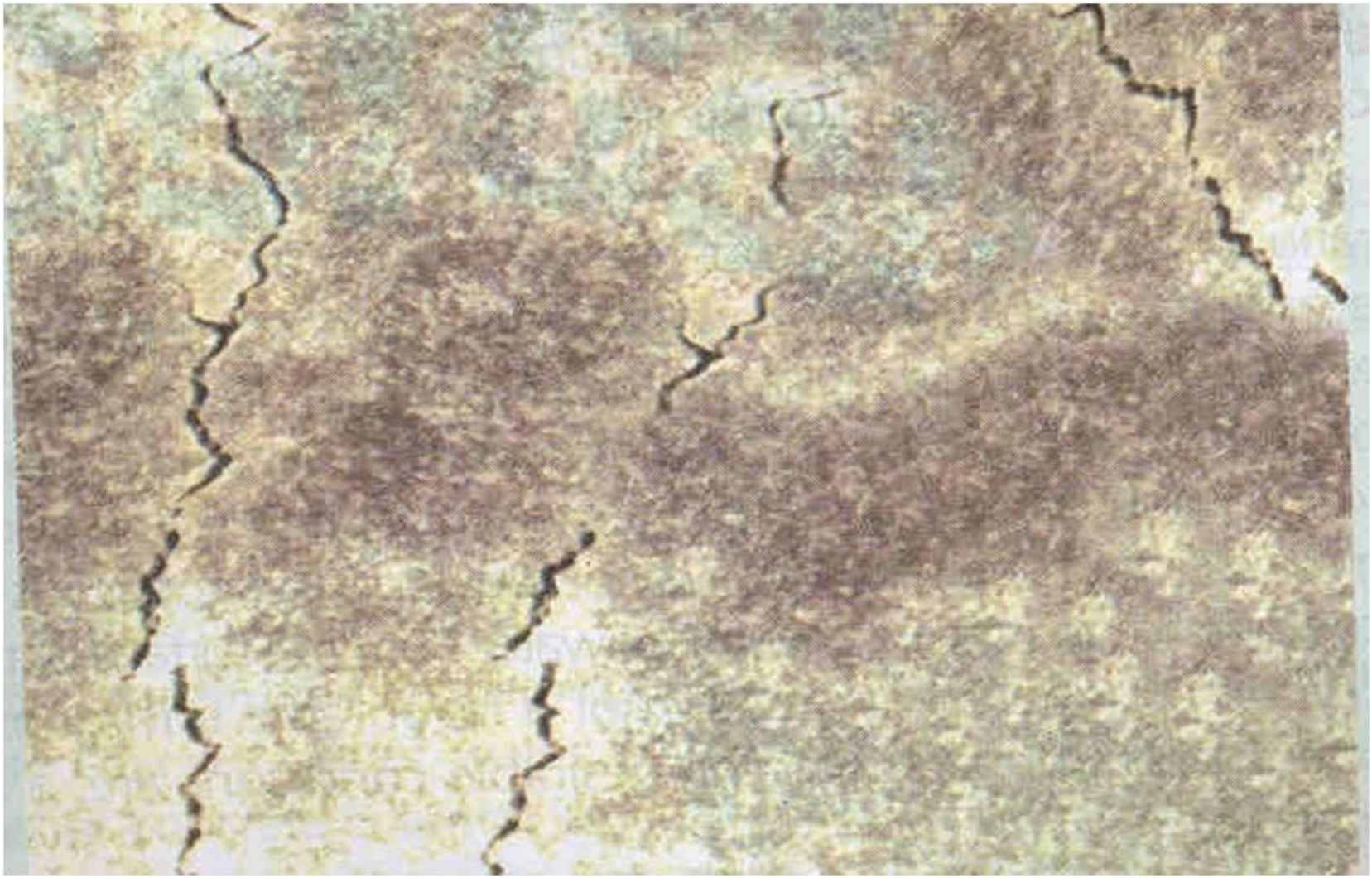
- Extremely important if the water-cement ratio is low & cement content is high.
- If the cement has a high rate of strength development
- Ambient temperature during placing is high and day is windy
- If the concrete contains GGBFS or Pulverized Fly ash (PFA)

Recommended Moist Curing (IS 456: 2000)

- **Normal Weather Conditions**
  - * OPC Based Concretes - 7 Days
  - * Blended Cements based concretes - 10 Days

- **Harsh Weather Conditions (Hot and Dry)**
  - * OPC Based Concretes - 10 Days
  - * Blended Cements based concretes - 14 Days

Cracks on concrete surface due to inadequate curing



Environment can be classified as

a) **Physical** –
Temperature, Moisture, alternate wetting and drying, freezing and thawing

b) **Chemical** –
Acidic, gaseous, alkaline, corrosive

Physical Environment impacts

Temperature

- Significantly affects rate of hydration of cement.
- Leads to Plastic shrinkage cracks in fresh concrete
- Volume changes and cracking especially in mass concrete
- Spalling and disintegration of concrete at higher temperatures > 250ºC (300ºc as per A.M. Neville P- 387)
- Variation in ambient temperature causes secondary stresses in structures

Physical Environment impacts

Moisture

- Shrinkage on drying, consequent volume change and cracking
- Induces corrosion of steel
- Acts as carrier of chemicals inside the body of concrete
- Causes efflorescence and deposition of Ca(Cao+2H_2 0= $Ca(OH)_2$+$H_2$0 OH)2 on surface
- Seepage / Leakages cause inconvenience to occupants and deteriorates structures due to permeable concrete.

Physical Environment impacts

Alternate wetting and drying

- Causes secondary stresses in the structures
- Accelerates corrosion of steel and
- Chemical attack on concrete

Freezing and Thawing

- Leads to expansion of concrete and cracking
- Ice-melting salts cause erosion of concrete

Cover to Reinforcement (IS 456 – 2000)

Exposure Condition	Min Nominal cover (mm)
Mild	20
Moderate	30
Severe	45
Very Severe	50
Extreme	75

Note :

1)For main reinforcement upto 12 mm dia bar for mild exposures, the nominal cover may be reduced by 5 mm

2) Unless specified otherwise, actual concrete cover should not deviate from the required nominal cover by +10mm or 0mm

Effect of Other materials

- Mineral Oil – usually effects only fresh concrete in their hardening process (petrol, petroleum distillates etc)
- Organic Acid – have corrosive effect
- Vegetable & Animal oils & Fats – causes deterioration of concrete surfaces due to their corrosive action
- Action of Sugar – has retarding effect on fresh concrete; has gradual corrosive effect on hardened concrete
- Action of Sewage – concrete sewers running full remain unaffected; but in partially filled sewers where hydrogen sulphide gas is evolved & sulphuric acid is formed, concrete above sewage level gets affected due to corrosive action of such acids

Why is Quality necessary?

- Intense Competitive Environment
- Entry of Foreign Goods and Services
- Increased Customer Consciousness
- Earning Profit instead of making profit
- Organization's survival / progress depends on it

- **Safety vs. Operation**
- **Quality vs. Progress**

Typical example in Railways

Operation vs safety



Both are PHODs/CHODs…

With different duties

and responsibilities

I am CSO

I am COM



In our day to day work COM gets a lot of importance…….Safety comes after Operation.

Similarly Quality comes after Progress..

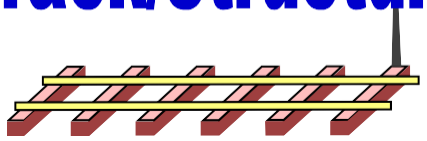
A KEY TO QUALITY WORK

Similarities between Human Body and Railway Track/Bridge/Structure



Man

Track/Structure



Segment



- Good Physique
- Regular Check ups
- Preventive Measures & Medicines
- Proper & Timely Cure & Restoration

LONG HEALTHY LIFE

- Good Construction
- Regular Inspections
- Preventive Measures & Materials
- Proper & Timely Repairs & Restoration

SMOOTH,SAFE TRAIN OPERATION & LONGER LIFE





This ROB constructed 14years back, was partly closed in 2007…





Distress in Concrete Slab at Bagha jatin ROB

GHPR Swimming Pool



Poor Quality & incomplete work by RITES as like this.

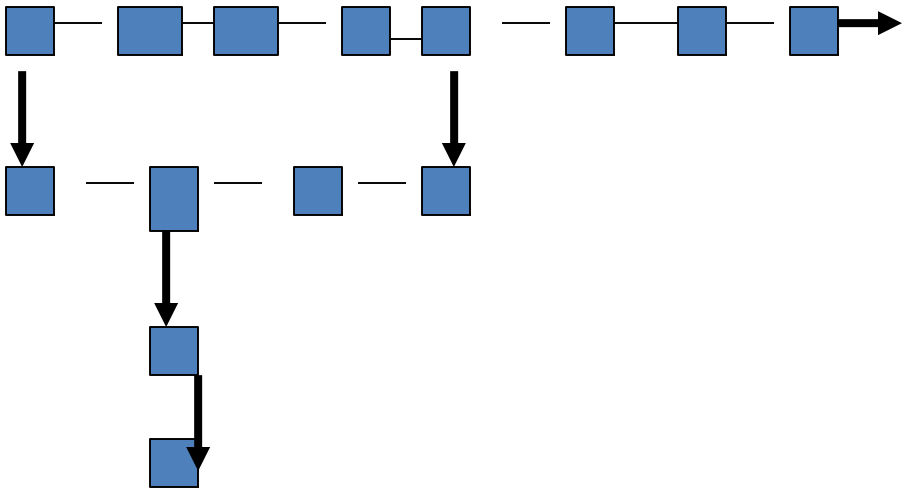


After construction & commissioning

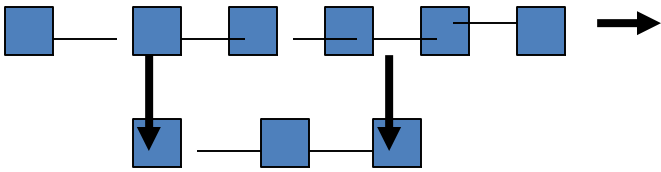


Construction Process Analysis

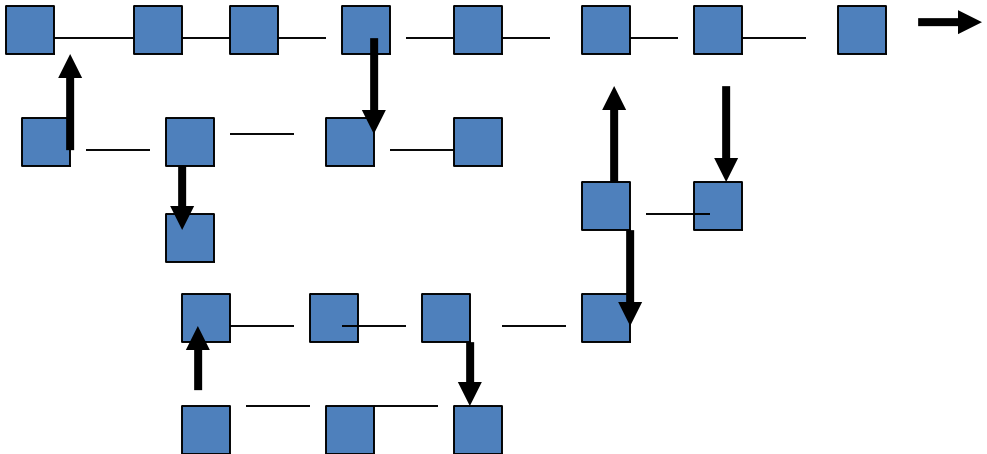
What we think it is



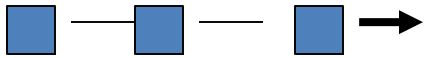
What it should be



What it really is (Hidden)



What it could be



5 **Gaps** in Service Quality

- **Gap 1:** Consumer Expectation- Management Perception
- **Gap 2:** **Management Perception-SQ Specification**
- **Gap 3:** SQ Specification- Service Delivery
- **Gap 4:** **Service Delivery- Perceived Service**
- **Gap 5:** Perceived Service- Consumer Expectation

5-S . . . . . .

- **Seiri** - Remove unnecessary items
- **Seiton** - Keep necessary things properly
- **Seiso** - Clean workplace thoroughly
- **Seiketsu** - Maintain high standards regularly
- **Shitsuke** - Maintain self-discipline